

Software Requirements Specification & Technical Blueprint

All-in-One Trade Ecosystem for Uganda and Africa
Working name: **SokoFlow** (placeholder — see §8.1 name availability)
Version 1.0 — 2 October 2026 — Prepared for DeryCode Technologies

Working name status: The name **SokoFlow** is **NOT** available as a clean brand. sokoflow.com is an active Kenyan FMCG sales-automation company (web, Google Play app, LinkedIn, X). The .ug and .africa domains are unregistered. The platform needs a different product name before any public launch or domain purchase. This document uses "the Platform" as the neutral working term.

1. Introduction

1.1 Purpose

This document specifies the full functional scope, non-functional requirements and technical architecture of the Platform: a unified trade ecosystem enabling consumer retail (B2C), person-to-person classifieds (C2C), wholesale and procurement (B2B/B2B2C), services and bookings, auctions, rentals, barter, dropshipping, agricultural trade, cross-border import ("Ship From Abroad"), institutional procurement (B2G) and digital commerce — all built on one shared Trade Engine, one account model and one financial ledger.

1.2 Scope

In scope: web app (Next.js), mobile app (Expo), seller portal, business portal, admin portal, partner interfaces, a Unified Trade API, the engine set listed in §3.2, and integration with PesaPal (MTN MoMo, Airtel Money, cards), couriers, freight/consolidation partners and SMS gateways.

Out of scope for v1.0 of this document: legal entity formation, marketing strategy, vendor contracts with couriers and freight partners, and licensed financial services (wallet balances held longer than order escrow requires, credit, insurance) — these are flagged in §8 as business workstreams.

1.3 Definitions

Term	Meaning
Trade	Any exchange of value: sale, wholesale purchase, service booking, auction win, rental, barter agreement.
Listing	A publishable offer of any type: product, used good, wholesale lot, service, rental item, auction lot, crop lot.
Transaction	The money-movement record backing a trade; always represented by ledger entries.
Escrow	Funds held by the Platform until delivery confirmation or dispute resolution releases them.
Landed cost	Item price + international freight + customs duty + VAT + local delivery, shown to the buyer before payment.
Workspace	A role-scoped environment inside one account (personal, business, supplier, farm, institution).

2. Overall Description

2.1 Personas and roles

One human = one account. Roles are capabilities activated inside that account, gated by verification level. A single person can hold every role simultaneously.

Role	Capabilities	Verification gate
Buyer	Purchase retail, bid, book services, request quotes, import goods	Phone (OTP)
Individual seller	List used goods (Deals), negotiate via chat, accept offers	Phone + national ID
Business owner	Retail storefront (Market), buy wholesale, list equipment for auction/rental, procure services, staff sub-accounts	Phone + ID + trade license
Supplier / Wholesaler	Publish wholesale lots with MOQ and tiered pricing, respond to RFQs, issue quotations	Phone + ID + trade license (+ tax ID)
Service provider	Publish bookable services with availability calendar	Phone + ID (+ license for regulated trades)

Farmer	List crop/livestock lots with harvest data, sell to bulk buyers	Phone (+ farm location)
Procurement officer	Create RFQs, manage purchase orders, approvals, vendor shortlists (B2B/B2G)	Business/institution workspace membership + role assignment

2.2 Unified account model

- FR-ACC-01** One identity per human (phone-first, email optional), one wallet, one ratings profile, one trust score.
- FR-ACC-02** A user creates/attaches workspaces; each workspace has its own storefront, orders, balances and staff permissions.
- FR-ACC-03** Role activation is progressive: the UI offers "Start selling", "Register a business", "Sell wholesale", "Offer services", "List farm produce", "Procure for my organization" entry points; each triggers its verification flow.
- FR-ACC-04** Users switch workspaces in one tap; all portals filter by active workspace.
- FR-ACC-05** Staff sub-accounts (cashier, manager, procurement officer) inherit scoped permissions of the parent workspace.

2.3 Assumptions and constraints

Primary launch market: Uganda, then East Africa. Connectivity is unreliable and metered; SMS and offline-conscious flows are first-class. Payments go through PesaPal (MoMo, Airtel Money, cards, cash-on-delivery). The Platform does not itself become a licensed financial institution in Phase 1; escrow is order-scoped, not a stored-value wallet product.

3. Technical Blueprint

3.1 Architectural style

Modular monolith with enforced module boundaries, one shared transaction model, centralized authorization, and a reliable double-entry ledger. Rationale: a small team can deliver the full 33-feature scope progressively without distributed-systems overhead; module boundaries and an internal event bus allow extracting individual engines into services later if scale demands it.

- One deployable API application (the monolith) exposing the Unified Trade API.
- Each engine is a module: owns its tables, may not read other modules' tables directly — only through published module interfaces or events.
- Background work uses a Postgres-backed job queue (pg-boss) and a transactional outbox table that doubles as the event log.

3.2 Component map

Interfaces	Next.js web app Expo mobile Seller portal Business portal Admin portal Partner API Unified Trade API (authn/authz, business rules, workflows, event processing, idempotency, rate limits)
Engines (modules)	Market Bulk & Procurement Deals & Offers Services & Booking Auction & Rental Agriculture Global Trade Finance & Ledger Logistics Trust & Safety Communication Analytics
Infrastructure	Supabase/PostgreSQL Object storage Realtime pg-boss jobs Event log (outbox) Reporting/rollups
External	PesaPal (MoMo/Airtel/cards) Couriers SMS gateway Freight/consolidation partners Banks Messaging

3.3 Technology stack

Layer	Choice	Notes
Web	Next.js (App Router, TypeScript)	Responsive, mobile bottom navigation, PWA caching for low bandwidth
Mobile	React Native / Expo (EAS builds)	One codebase for buyer and seller apps; deep links to WhatsApp
API	Node/TypeScript modular monolith	Deployed as serverless functions or container; OpenAPI-documented
Data	Supabase: PostgreSQL, Auth (with OTP),	Row-level security for workspace scoping; Postgres RLS as

	Storage, Realtime	defense-in-depth
Jobs/Events	pg-boss queue + transactional outbox	At-least-once delivery; all handlers idempotent
Payments	PesaPal	Same integration pattern as existing DeryCode products; IPN webhook verified
Messaging	In-app chat over Supabase Realtime; SMS fallback	WhatsApp share/deep links everywhere
Admin	Admin portal (Next.js)	Verification queue, disputes, duty rates, categories, boosts, moderation

3.4 Transaction and event model

FR-CORE-01 Every trade moves through one lifecycle state machine: created → reserved → paid (escrow) → fulfilled → released (or refunded/disputed/cancelled). Engine-specific flows (auction, rental, RFQ) extend, never bypass, this machine.

FR-CORE-02 All money mutations are double-entry ledger postings; no balance is stored as a mutable number — balances are derived from ledger sums.

FR-CORE-03 Every external write (payment charge, refund, courier dispatch, SMS) carries a client-supplied idempotency key; retries never duplicate.

FR-CORE-04 State changes emit domain events (OrderPlaced, PaymentReleased, ShipmentDelivered...) to the outbox; engines subscribe; an event viewer in admin shows the log.

3.5 Authorization model

Centralized policy layer: role + workspace + verification level + listing ownership checks in one place, applied by every engine. Postgres RLS mirrors workspace scoping as a second barrier. Admin actions are separately permissioned and audit-logged.

4. Functional Requirements by Engine

Requirement IDs are stable and referenced by the roadmap (§7) and test plans. Priority: all Phase 1-2 items are P0; later phases inherit priority from their release phase.

4.1 Market Engine (B2C retail)

FR-MKT-01 Business storefronts: profile, banner, policies, ratings.

FR-MKT-02 Product listings: images, variants (size/color), price, stock, category tree (admin-managed).

FR-MKT-03 Cart and checkout with MoMo/Airtel/card/COD; escrow by default.

FR-MKT-04 Promotions: coupons, flash sales, featured placement (paid boost).

FR-MKT-05 Delivery: courier integration or station pickup; buyer confirms delivery to release escrow.

FR-MKT-06 Returns: policy windows per category; refund flows post back through the ledger.

4.2 Deals & Offers Engine (C2C classifieds)

FR-DEAL-01 Free listings with expiry and easy repost.

FR-DEAL-02 Chat with structured offers (amount, meet/deliver), accept/counter.

FR-DEAL-03 Optional escrow for shipped deals; pure meet-ups settle in cash.

FR-DEAL-04 Boosted/promoted listings (daily/weekly fees) with clear ad labeling.

FR-DEAL-05 Safety: report/block, number masking, verified-ID seller badge.

4.3 Bulk & Procurement Engine (B2B/B2B2B)

FR-BULK-01 Wholesale lots: MOQ, quantity-tiered pricing, lead time, sample orders.

FR-BULK-02 RFQ: buyer posts request (specs, quantity, deadline); matching suppliers respond with quotations.

FR-BULK-03 Quotation → Purchase Order → fulfillment milestones → escrow release per milestone.

FR-BULK-04 Supplier profiles with verified-supplier badge, response-rate stats, certifications.

FR-BULK-05 Business buyer dashboards: recurring orders, approved-vendor lists, approval chains for POs (procurement officer role).

FR-BULK-06 Contracts & subscriptions: standing prices and scheduled deliveries between verified parties.

4.4 Services & Booking Engine

FR-SVC-01 Service listings with fixed or hourly pricing, service area, calendar availability.

FR-SVC-02 Booking flow with deposit (escrow) and balance on completion.

FR-SVC-03 Reviews specific to services; repeat-booking discounts.

4.5 Auction & Rental Engine

FR-AUC-01 Auction lots: reserve price, bid increments, auto-extend on late bids, winner escrow.

FR-RENT-01 Rental listings: day/week/month rates, security deposit, check-in/check-out condition photos, late-return fees.

FR-RENT-02 Machinery/equipment rental with operator add-on option (Ugandan machinery market).

4.6 Agriculture Engine

FR-AGR-01 Crop/livestock lots: quantity, grade, harvest date, location; bulk-buyer browsing by region.

FR-AGR-02 Market-price reference board (admin-managed weekly prices) shown to farmers.

FR-AGR-03 Verified buyer program for produce purchases (reduces fear of fake buyers); escrow release on delivery at agreed grade.

4.7 Global Trade Engine (Ship From Abroad)

FR-GLB-01 Import listings: item URL or catalog from partner suppliers (China, UAE, US, UK), or buyer-submitted link ("Buy For Me").

FR-GLB-02 Landed-cost calculator shown pre-payment: item + freight estimate + duty + VAT + local delivery, using admin-managed duty-rate tables per HS category.

FR-GLB-03 Consolidation-hub flow: items arrive at hub (e.g. Guangzhou/Dubai) → consolidated shipment → customs clearance → local delivery; buyer sees status at every hop.

FR-GLB-04 Escrow staged: deposit at order, balance on arrival at hub or per-milestone, refund policy for lost/seized goods clearly displayed.

FR-GLB-05 Multi-currency display (USD/CNY source prices) with UGX settlement via PesaPal.

4.8 Finance & Ledger Engine

FR-FIN-01 Double-entry ledger accounts per workspace: escrow-held, available, commission-owed, refunds.

FR-FIN-02 Commission rules per engine/category (percent or fixed); automatic split at escrow release.

FR-FIN-03 Payouts to MoMo/bank; payout batch review in admin; holds and reserves policy.

FR-FIN-04 Reconciliation: daily PesaPal settlement vs ledger; mismatch alerts.

FR-FIN-05 Traceable refunds: every refund references the original transaction and its ledger entries.

FR-FIN-06 All amounts integer minor units (UGX has no minor unit; USD cents for imports).

4.9 Logistics Engine

FR-LOG-01 Courier integrations (API partners + manual dispatch fallback); pickup/drop-off stations.

FR-LOG-02 Tracking timeline surfaced to buyer and seller in-app, with SMS on key hops.

FR-LOG-03 Delivery confirmation (buyer tap, courier confirm, or station pickup code) triggers escrow release.

FR-LOG-04 Zone-based delivery pricing table managed by admin; used by all engines.

4.10 Trust & Safety Engine

FR-TRU-01 Verification tiers: phone → ID → business license → supplier/verified badges; each tier unlocks roles (§2.1).

FR-TRU-02 Disputes: evidence (photos, chat), resolution window, admin arbitration, outcomes post to ledger.

FR-TRU-03 Fraud monitoring: velocity rules, listing-scanning moderation queue, device/account risk signals.

FR-TRU-04 Ratings and reviews per role-relationship (buyer, seller, supplier, service provider), with response right.

FR-TRU-05 Buyer protection guarantee: clear eligibility rules published per mode.

4.11 Communication Engine

FR-COM-01 Realtime chat (Supabase Realtime) with image messages, offer cards, and block/report.

FR-COM-02 Notification matrix: push, SMS (order events, delivery hops), email optional.

FR-COM-03 WhatsApp deep links for sharing listings and order updates.

FR-COM-04 Language templates: English, Luganda, Runyoro, Swahili.

4.12 Analytics Engine

FR-ANA-01 Seller dashboards: views, conversion, stock alerts, payout history.

FR-ANA-02 Business dashboards: procurement spend, top vendors, PO cycle time.

FR-ANA-03 Admin: GMV, take rate, dispute rate, verification funnel, category heatmaps; all from rollup tables, not live scans.

FR-ANA-04 Trade intelligence (Phase 6): price trends, demand forecasting, sourcing suggestions.

5. Non-Functional Requirements

Area	Requirement
Security	OTP auth + optional PIN/biometric on mobile; RBAC everywhere; encrypted storage of ID documents; rate limits per endpoint; security audit before public launch; admin actions audit-logged.
Financial integrity	Double-entry ledger (FR-FIN-01); idempotent payment/refund handling (FR-CORE-03); daily reconciliation (FR-FIN-04); no floating balances; currency-safe integer math.
Reliability	At-least-once jobs with idempotent handlers; duplicate-order protection via idempotency keys; daily backups + tested restore; event-log monitoring dashboards.
Regulatory readiness	Business registration, consumer-protection and returns policy, privacy policy and data retention, tax (VAT) handling on fees, URA customs data sources for duty tables, payments compliance via PesaPal; per-market country config.
Localization	English, Luganda, Runyoro, Swahili at launch; UGX default, multi-currency display; low-bandwidth UI (image compression, PWA caching); SMS for feature phones.
Extensibility	OpenAPI-documented Unified Trade API; partner API tier; configurable workflows (approval chains, policy windows, duty tables) without code changes.
Accessibility	Mobile-first interfaces; large-touch targets; screen-reader basics on web; plain-language flows; assisted onboarding for low digital literacy.
Performance	Category/search pages < 2.5s LCP on 3G-simulated profiles; listing pages static-cached; search via Postgres FTS at launch (dedicated search engine later).

6. Data Model Overview (core entity groups)

Group	Key tables
Identity	users, profiles, workspaces, workspace_members, verifications, roles/permissions
Catalog	listings (polymorphic: product, deal, wholesale_lot, service, rental, auction, crop_lot), variants, media, categories, locations
Trade	orders, order_items, rfqs, quotations, purchase_orders, bookings, bids, rental_agreements, barter_offers, subscriptions
Finance	ledger_accounts, ledger_entries, transactions, payment_attempts, payouts, refunds, commission_rules, wallets (derived)
Fulfilment	shipments, shipment_events, delivery_zones, rates, hubs, customs_declarations
Trust	disputes, evidence, reviews, sanctions, risk_signals, moderation_queue
Comms	conversations, messages, notifications, sms_log, language_templates
Ops	outbox_events, jobs, audit_log, settings, duty_rates, country_config

The detailed schema with column types, constraints and RLS policies is the next deliverable after this document is approved.

7. Release Roadmap (6 phases, 33 features)

Phase 1 — Core Trade Infrastructure (6 features, exit: first real order end-to-end)

Unified accounts, identities and roles · Trade Engine and transaction lifecycle · Catalog, listings, pricing and inventory · Payment integration and financial ledger · Order management, delivery and refunds · Trust, verification, disputes and admin.

Phase 2 — Market and Deals (5, exit: public B2C + C2C launch in Kampala/Kyenjojo)

B2C retail marketplace · C2C classifieds · Chat, offers and negotiation · Ratings, returns and promoted listings · Mobile app and offline-conscious workflows.

Phase 3 — Business and Wholesale (5, exit: first B2B PO completed with escrow milestones)

B2B and B2B2B wholesale · C2B and D2C commerce · RFQs, quotations and purchase orders · Subscriptions, contracts and procurement · Business dashboards and accounting integrations.

Phase 4 — Extended Trade (6, exit: two new trade modes live with real transactions)

Services marketplace and bookings · Agricultural trade · Auctions and bidding · Rental and leasing · Barter and consignment · Dropshipping and recurring orders.

Phase 5 — Global and Institutional Trade (5, exit: first consolidated shipment cleared and delivered)

Import and export management · Ship From Abroad and consolidation hubs · Customs, freight and landed-cost calculations · B2G procurement and institutional workflows · Multi-country and multi-currency support.

Phase 6 — Advanced Trade Ecosystem (6, exit: partner APIs in external use)

Trade intelligence and forecasting · Advanced business reporting · Loyalty, referrals and affiliate tools · Digital commerce · Financial service integrations (subject to approval) · Partner APIs and ecosystem integrations.

8. Findings, Risks and Open Questions

8.1 Working name availability — SokoFlow is NOT available

sokoflow.com is an active Kenyan sales-automation company (FMCG/distribution) with a Google Play app and company presence on LinkedIn and X. Using "SokoFlow" would create brand confusion in the same region and sector. sokoflow.ug and sokoflow.africa are unregistered, but the domain being free does not fix the brand conflict. **Decision needed:** choose a different name before Phase 2's public launch. Candidate directions: trade/market-themed names in Swahili/Lunyoro that remain pronounceable in English; availability will be checked for the chosen shortlist the same way (web, app stores, domains).

8.2 Key risks

1. Scope breadth: 33 features across 6 phases — mitigated by the modular monolith and strict phase exits.
2. Escrow perception: users must trust the Platform holds funds; mitigated by buyer-protection guarantees, clear policy, and PesaPal's familiar flow.
3. Ship From Abroad customs complexity: duty tables and clearance partners need real operational agreements before Phase 5 ships.
4. Regulatory: consumer protection, tax and (later) financial-services rules; legal review needed per market entry.
5. Fraud and disputes on C2C: Trust & Safety Engine must launch with Phase 2, not after.

8.3 Open questions for the owner

1. Final product name (SokoFlow must be replaced).
2. Launch geography order after Uganda (Kenya? Tanzania? DRC?).
3. Initial commission percentages per mode.
4. Preferred hub cities for Ship From Abroad (Guangzhou? Dubai? both?).
5. Single Supabase project or separate projects for staging/production.

9. Next Deliverables

1. Database schema (DDL, constraints, RLS policies) for Phase 1 tables.
2. Unified Trade API design (OpenAPI) for Phase 1 endpoints.
3. Repository structure (monorepo: web, mobile, api, supabase, docs).
4. Phase 1 implementation: accounts/roles, Trade Engine lifecycle, catalog, PesaPal + ledger, orders/delivery/refunds, trust/admin.